Math 421

Wednesday, October 29

Announcements

- Grace's Drop-In Hours
  - **Today** 12-1 pm VV 311
  - Thursday 9-11 am VV B205
- MLC
  - Proof Table: M-Th 3-7:30 VV B227
  - Course Assistant: Th 4-7 Table 4
- Homework 7 Due Friday
- Exam 2 – 11/12 5:45-7:15 pm
  - Email me & fill out the form if you have a conflict and need to take the makeup.
  - Sign up for McBurney exam if applicable.

Convergent $\Leftrightarrow$ Cauchy

Theorem 10.11. A sequence is a convergent sequence if and only if it is a Cauchy sequence.

Proof. (Cauchy $\Rightarrow$ Convergent).

Chapter 2: Sequences

Section 14: Series

Cauchy Criterion

Def. We say a series $\sum a_n$ satisfies the Cauchy criterion if its sequence of partial sums is a Cauchy sequence.

Theorem 14.4. A series converges if and only if it satisfies the Cauchy criterion.

Activity – Prove using Theorem 14.4

Corollary 14.5. If a series $\sum a_n$ converges, then $\lim a_n = 0$.

Proof.

Comparison Test

Theorem 14.6 (i). Let $\sum a_n$ be a series where $a_n \geq 0$ for all n. If $\sum a_n$ converges and $|b_n| \leq a_n$ for all n, then $\sum b_n$ converges.

Proof.

Chapter 3: Continuity

Section 20:Limits of Functions

Functions

Def. Given two sets A and B, a **function** f from A to B is a mapping that takes each element $x \in A$ and associates with it a single element $f(x) \in B$. In this case we write $f: A \to B$.

- The set A is called the **domain of $\boldsymbol{f}$**. For us, $\mathrm{dom}(f) \subseteq \mathbb{R}$.
- The set $\{y \in B : y = f(x) \text{ for some } x \in A\}$ is called the **range of $\boldsymbol{f}$**.

Example: Define $f: \mathbb{R} \to \mathbb{R}$ by $f(x) = x^2$. Then $[0, \infty)$ is the range of f.

Activity – Find the natural domains

1. $f(x) = \frac{1}{x}$

2. $g(x) = \sqrt{4 - x^2}$

3. $\csc x = \frac{1}{\sin x}$

Functional Limits – Non-Rigorous

Non-rigorous definition: $\lim_{x \to a} f(x) = L$ means that the values of $f(x)$ are close to L when the values of x are close to (but not equal to) a.

Examples & Non-Examples.

Functional Limits – via Sequences

Def. Let $f: S \to \mathbb{R}$ and let a be the limit of some sequence in S.

Then $\lim_{x \to a} f(x) = L$ if for every sequence (x_n) in S with limit a, we have $\lim_{n \to \infty} f(x_n) = L$.

Example: Let $f(x) = x^2$, $\lim_{x \to 0} x^2 = 0$.

Activity

Prove carefully using sequences that

$$\lim_{x \to 3} x^2 - x = 6.$$

Functional Limits – via $\epsilon - \delta$

Def. Let $f: S \to \mathbb{R}$ and let a be the limit of some sequence in S.

Then $\lim_{x \to a} f(x) = L$ if for all $\epsilon > 0$ there exists some $\delta > 0$ such that:

$$\text{if } 0 < |x - a| < \delta \text{ and } x \in S, \text{ then } |f(x) - L| < \epsilon.$$

Example

Let $f(x) = \frac{1}{2}x$, $a = 2$, $L = 1$.

- If $0 < |x - 2| <$ ______, then $|f(x) - 1| < 1$.
- If $0 < |x - 2| <$ ______, then $|f(x) - 1| < 1/2$.

Example

Let $f(x) = \begin{cases} x+1, & x \le 1 \\ 2x, & x > 1 \end{cases}$, $a = 1$, $L = 2$.

- If $0 < |x - 1| <$ ______, then $|f(x) - 1| < 1$.
- If $0 < |x - 1| <$ ______, then $|f(x) - 1| < 1/2$.

Activity

Let $f(x) = \begin{cases} x - 1, & x \leq 2 \\ 7 - 3x, & x > 2 \end{cases}$

Fill in the blanks

- If $0 < |x - 2| <$ ______, then $|f(x) - 1| < 1$.
- If $0 < |x - 2| <$ ______, then $|f(x) - 1| < 1/2$.
- If $0 < |x - 2| <$ ______, then $|f(x) - 1| < \epsilon$.

$\epsilon - \delta$ Proof Recipe

To show that $\lim_{x \to a} f(x) = L$.

1. "Fix $\epsilon > 0$."
2. Determine δ, requires scratch work
3. "Let $\delta =$ _____"
4. "Assume $0 < |x - a| < \delta$."
5. Show that $|f(x) - L| < \epsilon$.
6. "Since $\epsilon > 0$ was arbitrary, $\lim_{x \to a} f(x) = L$."

Example

Show $\lim_{x \to 2} 6x = 12$.

Scratch work.

Proof.

Activity – Fill in the Blanks

Thm. $\lim_{x \to 2} 5x = 10$.

Proof. Fix $\epsilon > 0$. Let $\delta =$ __________. Assume $0 < |x - 2| < \delta$. Then

$|5x - 10| =$ ____

____ $< \epsilon$.

Since $\epsilon > 0$ was arbitrary, $\lim_{x \to 2} 5x = 10$.

Example – $\lim_{x \to 1} x^2 + x + 1 = 3$

Scratch work.

Algorithm for quadratic polynomials:

1. Factor out $|x - a|$ from $|f(x) - L|$
2. Deal with the extra factor
   a) Write in terms of $x - c$:
   b) Use triangle inequality:
   c) Assume a bound of 1:
3. Let $\delta = \min\{1, \epsilon/\text{answer in c)}\}$

Example – $\lim_{x \to 1} x^2 + x + 1 = 3$

Proof.

Activity – Fill in the Blanks

Thm. $\lim_{x \to 3} x^2 - x = 6.$

Proof. Fix $\epsilon > 0$. Let $\delta = \min\{1, \_\_\_\_\_\_\_\_\}$. Assume $0 < |x - 3| < \delta$.

Then $|x^2 - x - 6| = \_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_$

$\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_$

$\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_ < \epsilon.$

Since $\epsilon > 0$ was arbitrary, $\lim_{x \to 3} x^2 - x = 6.$

Activity – Fill in the Blanks

Thm. If $\lim_{x \to a} f(x) = L$, then $\lim_{x \to a} 4f(x) = 4L$.

Proof. Fix $\epsilon > 0$. Since $\lim_{x \to a} f(x) = L$, there exists $\delta > 0$ such that:

$$\text{if } 0 < |x - a| < \delta, \text{ then } |f(x) - L| < \_\_\_\_\_\_\_.$$

Assume $0 < |x - a| < \delta$. Then

$|4f(x) - 4L| =$ ____

____

____ $< \epsilon$.

Since $\epsilon > 0$ was arbitrary, $\lim_{x \to a} 4f(x) = 4L$.